

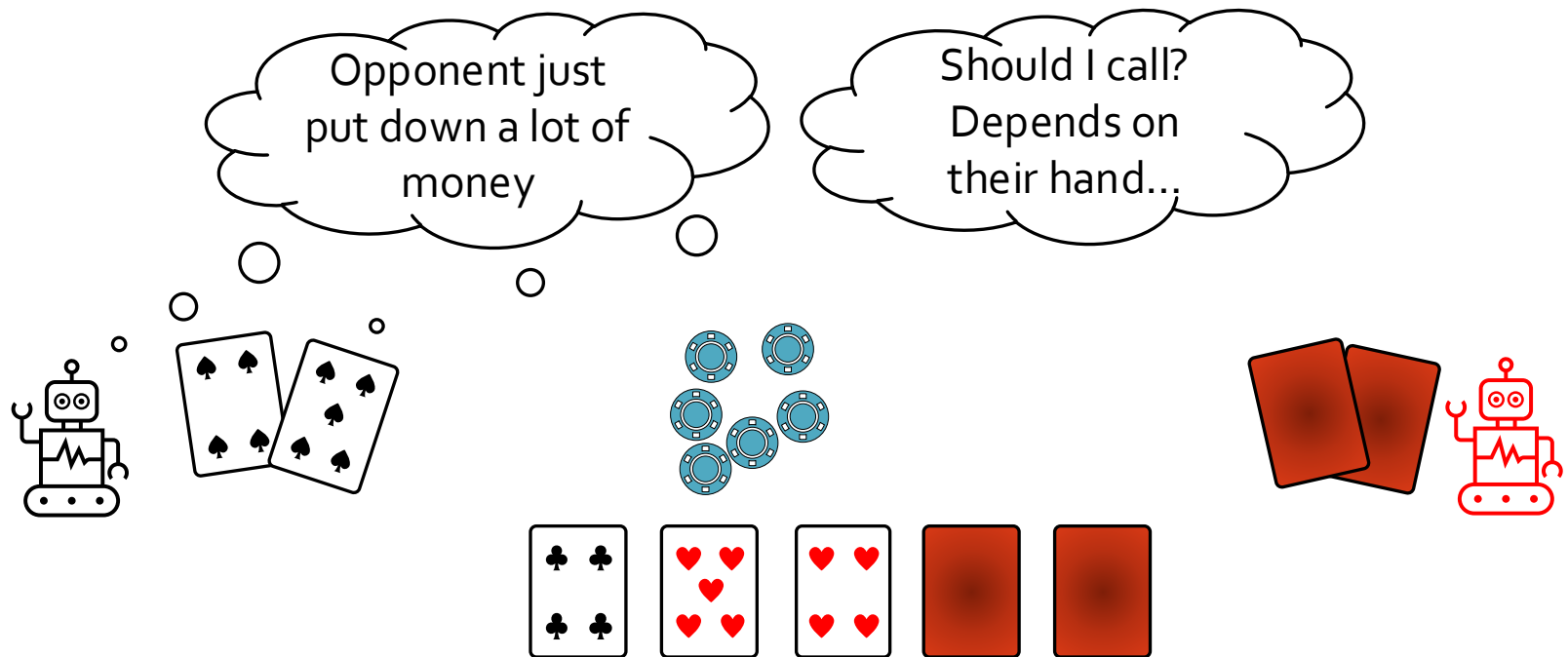


Partially Observable Games

AIMA Chapter 5.6

See also Osborne's "Introduction to Game Theory" Chapter 10

Information Sets



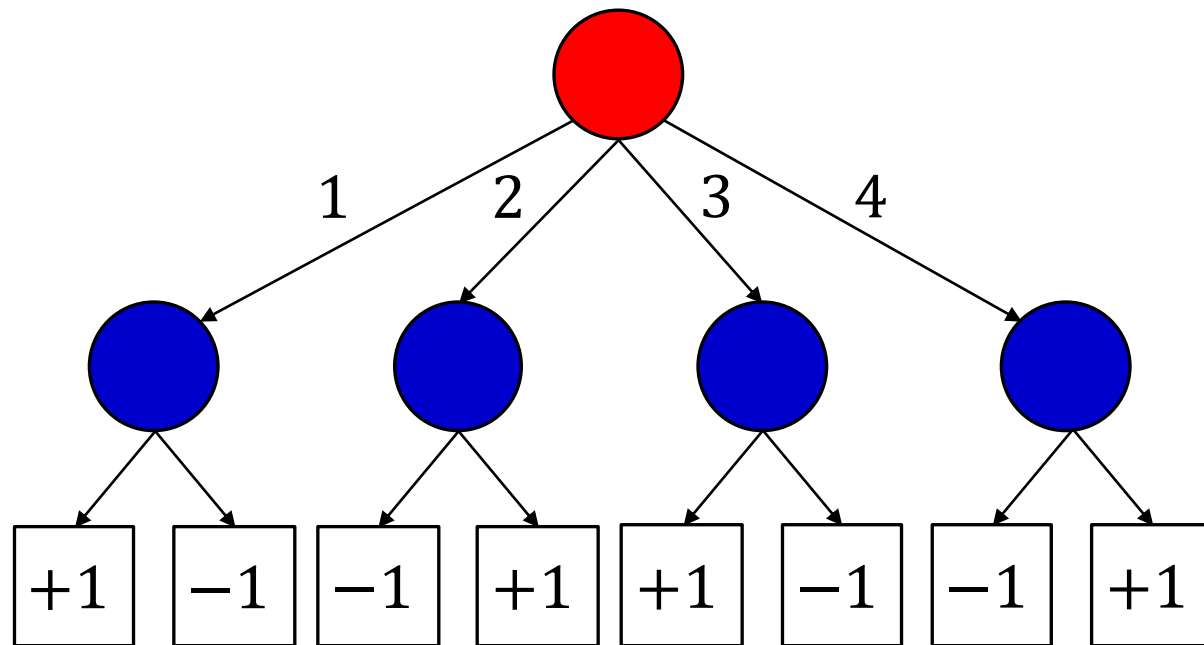
The information set is a union of all possible game states that we can be in (based on information available to us right now).



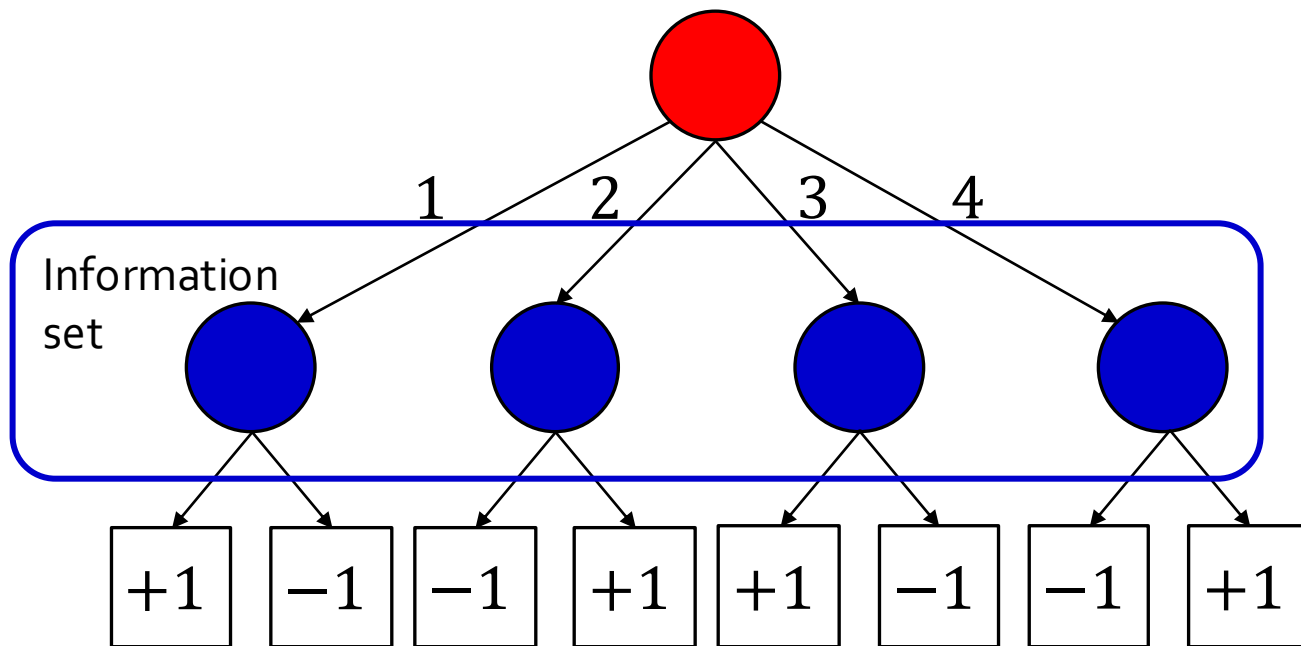
Information Sets

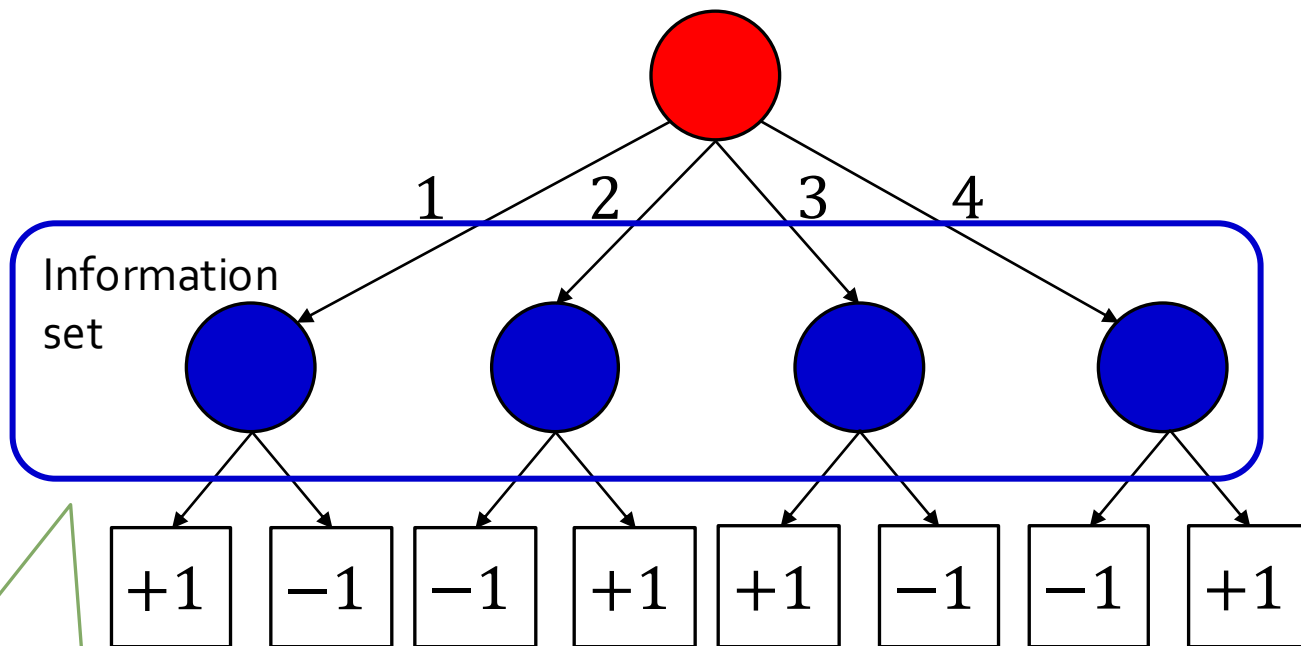
Player 1 picks a number from 1 to 4.

Player 2 guesses whether it is even or odd.



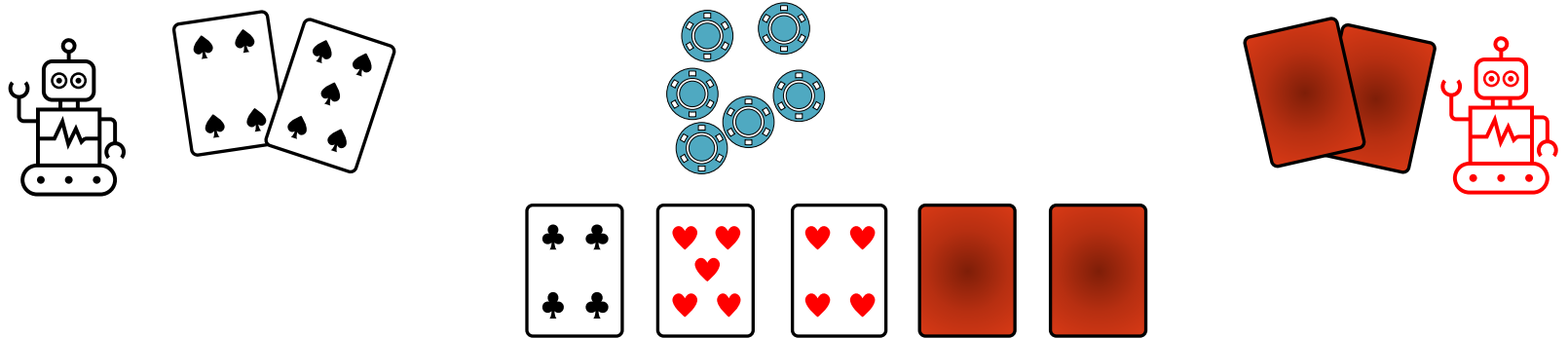
Right is guessing even, and left is guessing Odd





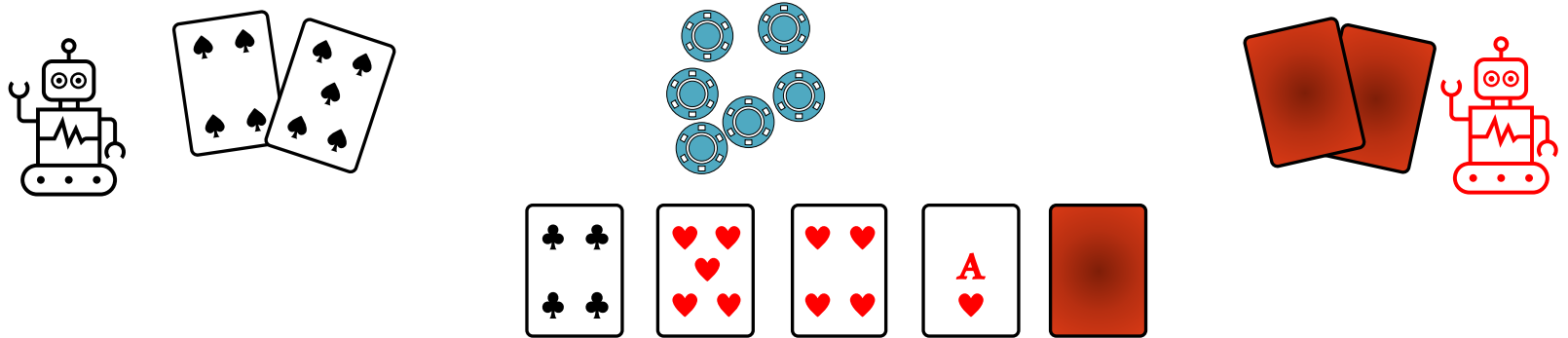
Player 2 cannot distinguish between these four states.

Information Sets



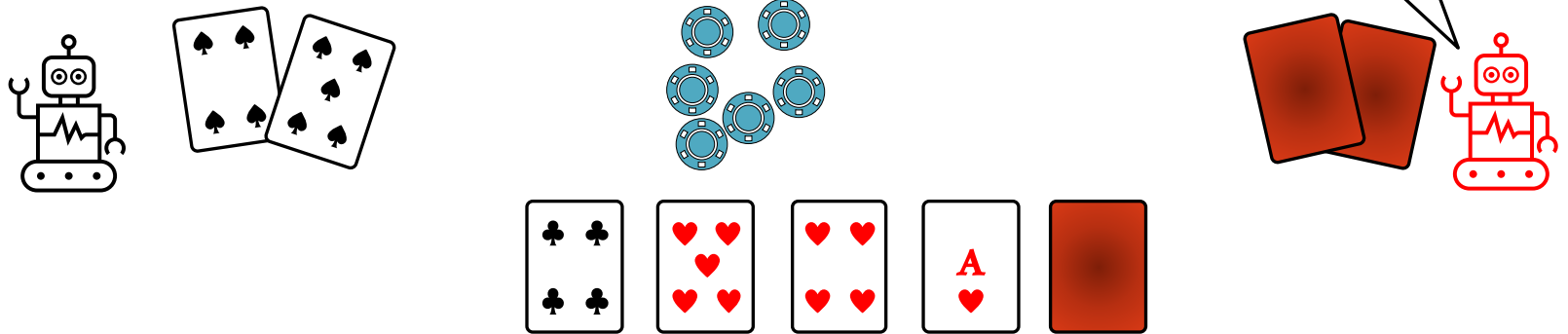
We can rule out certain states based on other players' behavior.

Information Sets



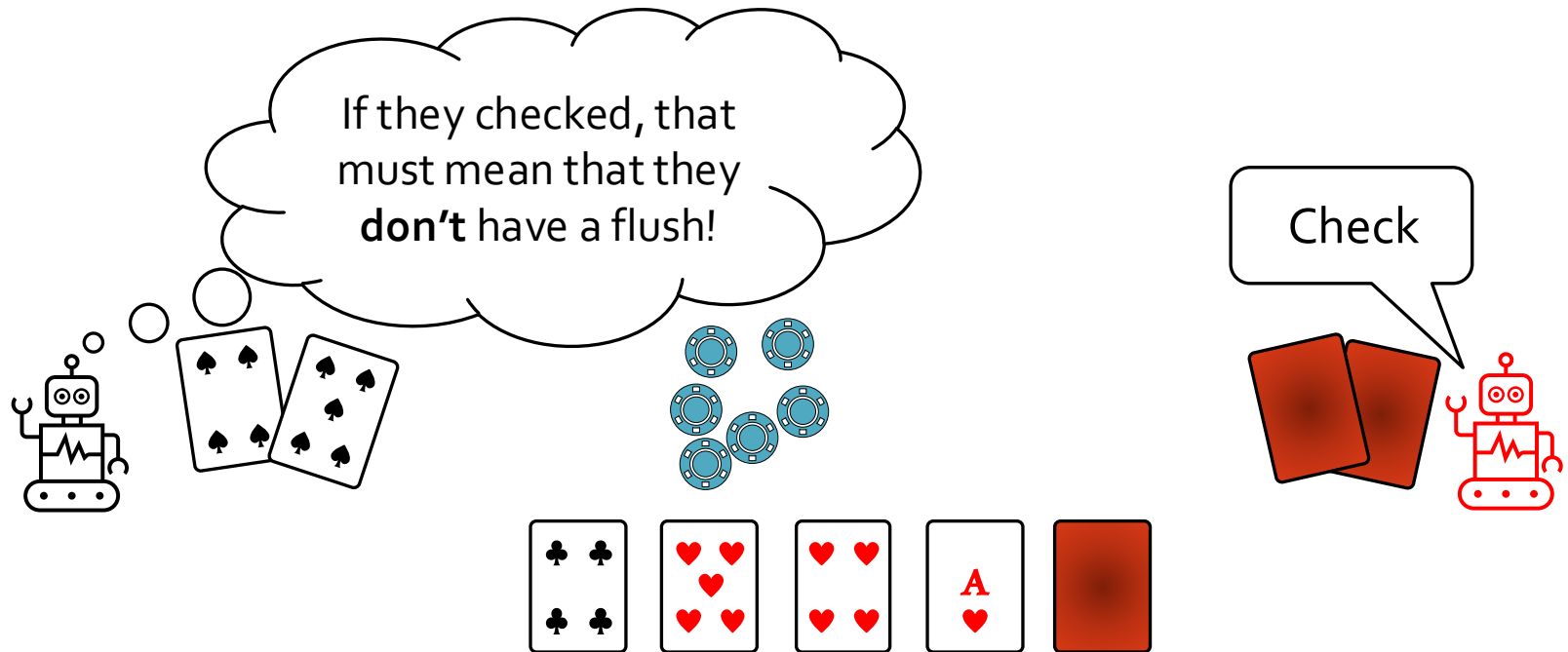
We can rule out certain states based on other players' behavior.

Information Sets



We can rule out certain states based on other players' behavior.

Information Sets



We can rule out certain states based on other players' behavior.

The Hat Game



Adversary puts red hats and blue hats on players' heads.

Players see other players' hats, but not their own.

At least one hat is red.

On their turn, a player says **red**, **blue** or pass.

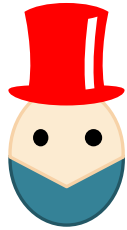
If all players guess their hat color correctly, they win, otherwise they lose.

The Hat Game



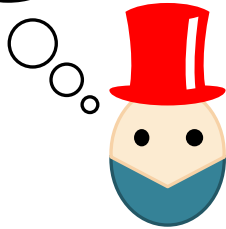
Take a minute to think of how this game can be solved.

The Hat Game



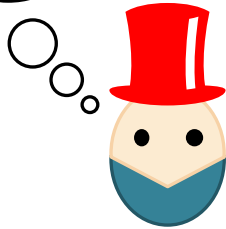
The Hat Game

At least one hat is
red, and player 2's
is blue so...



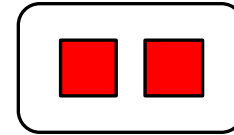
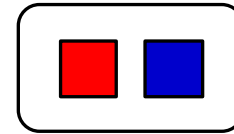
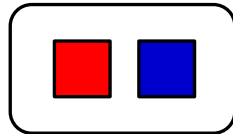
The Hat Game

At least one hat is
red, and player 2's
is blue so...

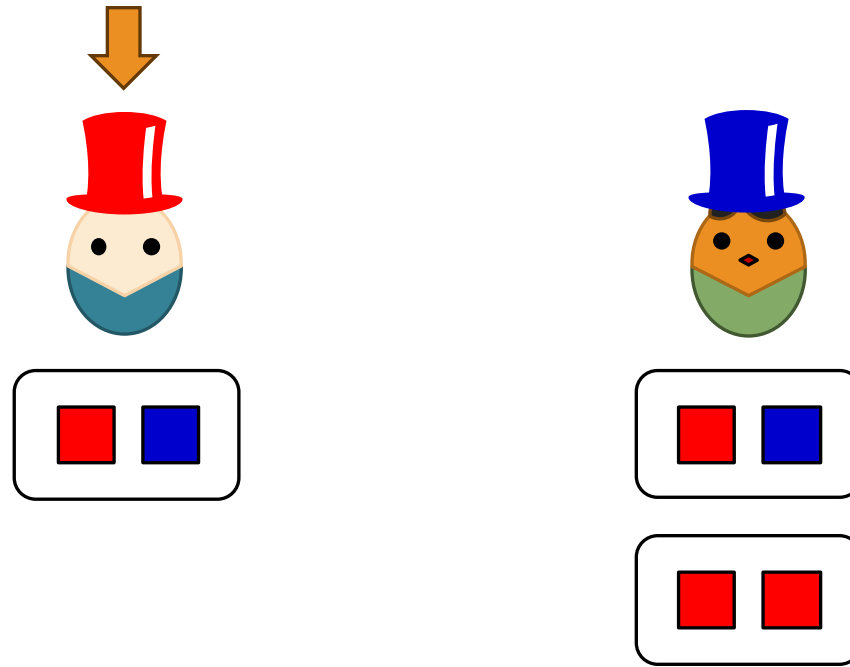


I have no idea
what's happening

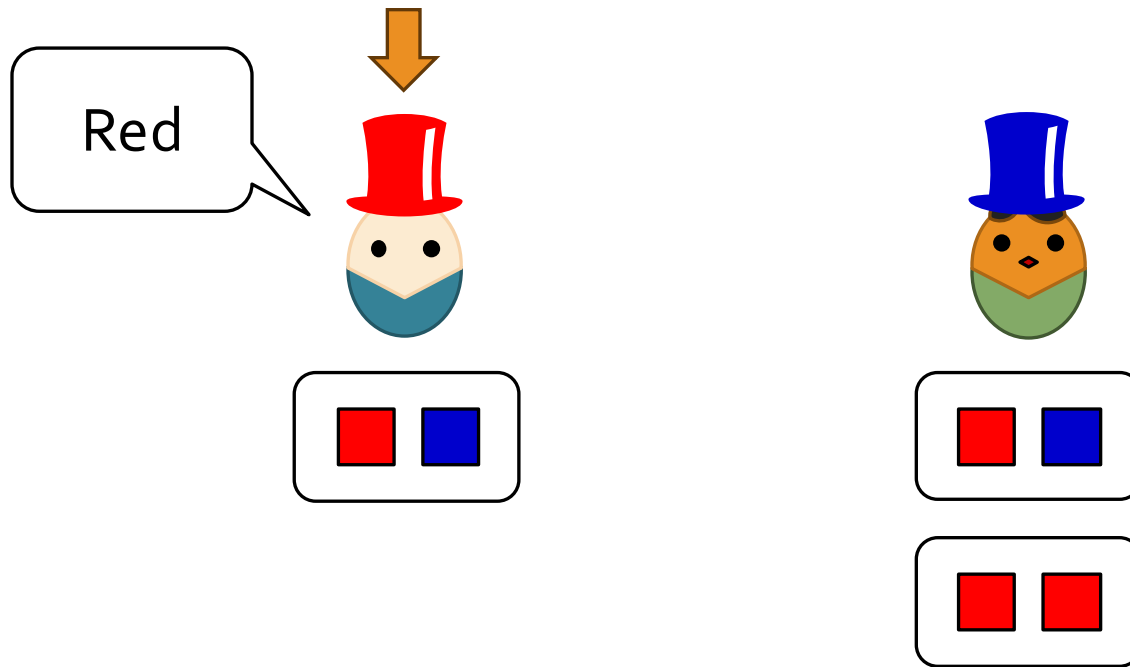
The Hat Game



The Hat Game

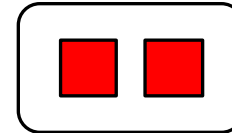
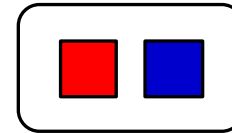
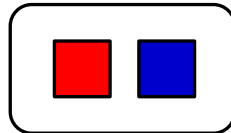


The Hat Game



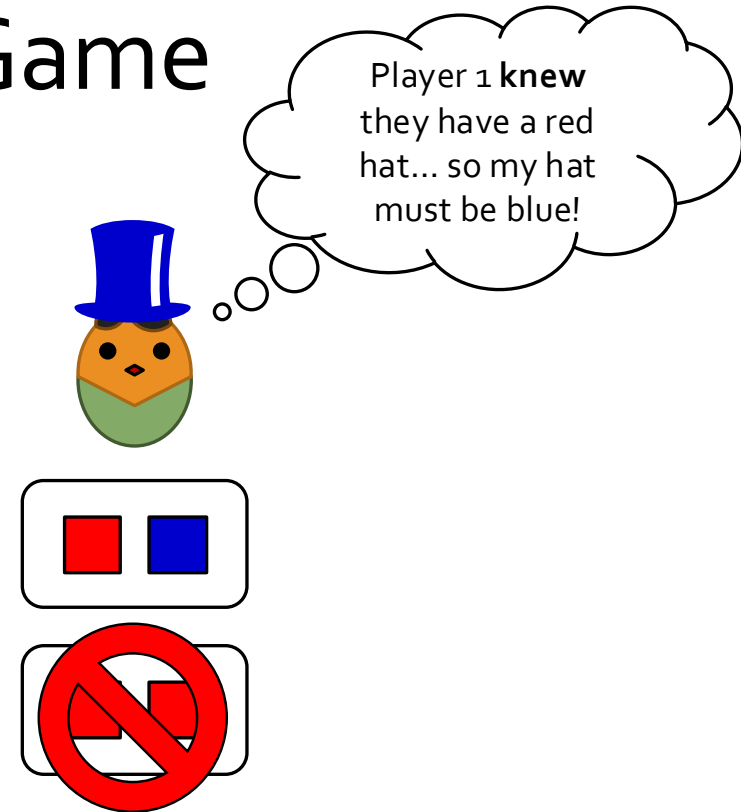
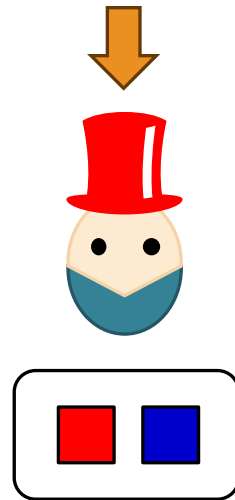
The Hat Game

Red

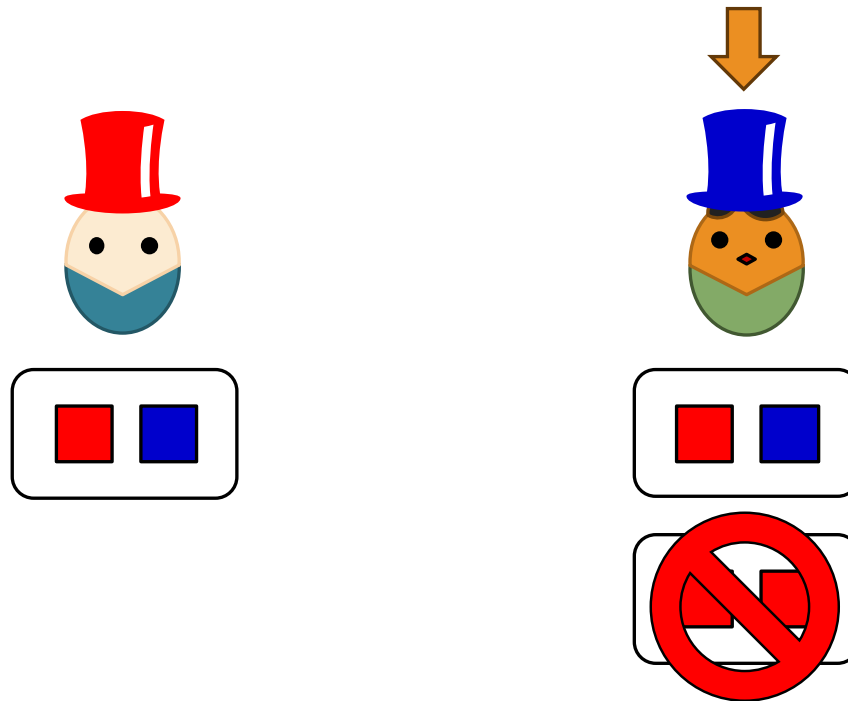


Player 1 **knew**
they have a red
hat... so my hat
must be blue!

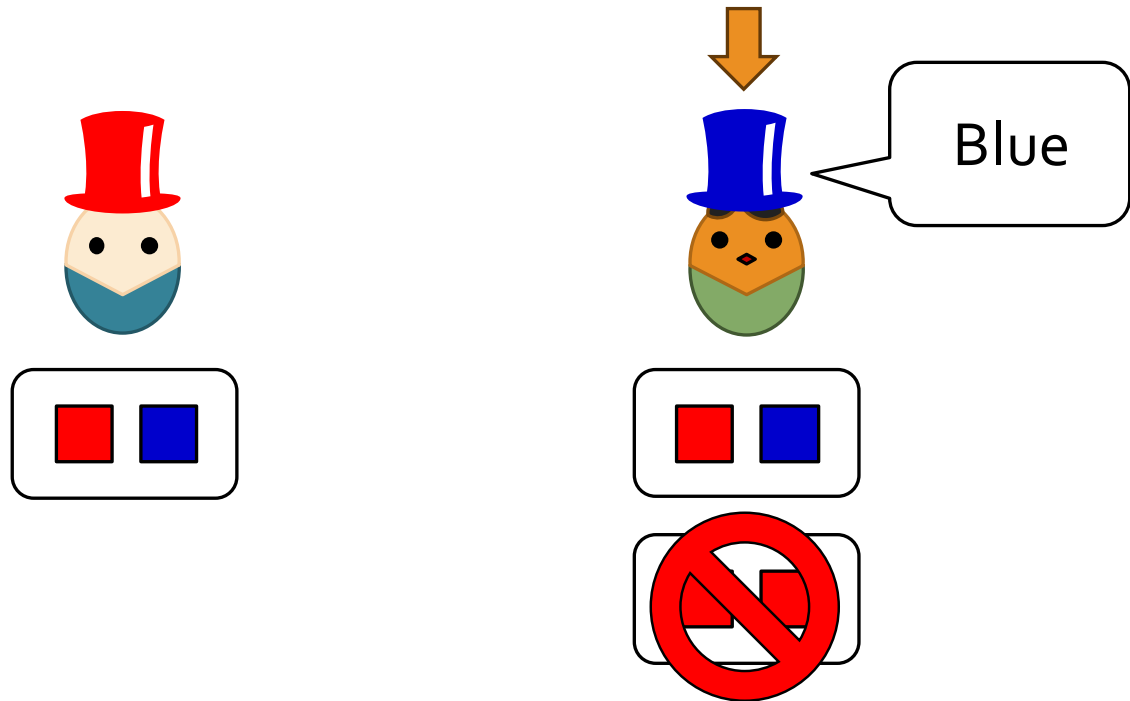
The Hat Game

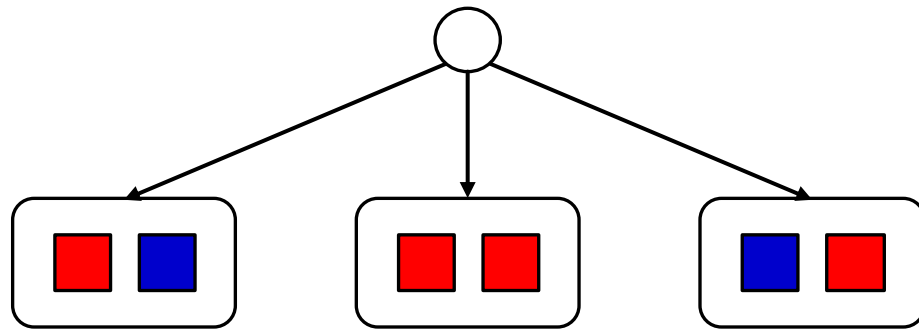


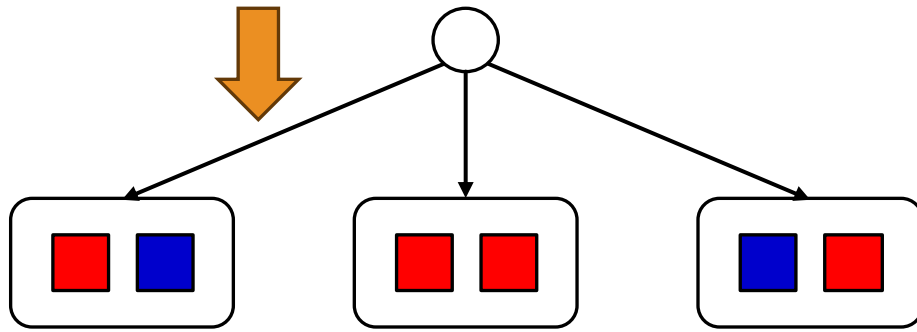
The Hat Game

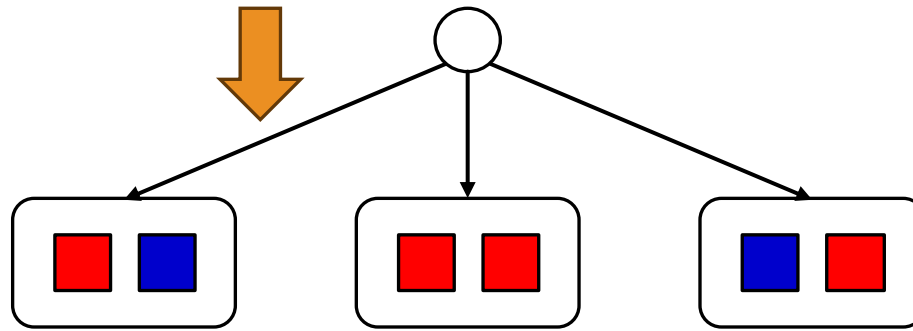


The Hat Game

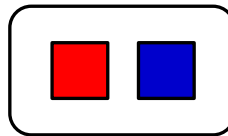




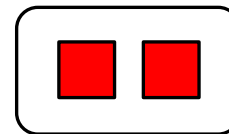
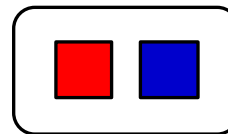




Player 1's
information set:



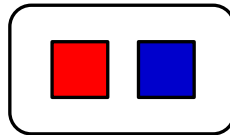
Player 2's
information set:



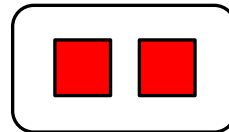
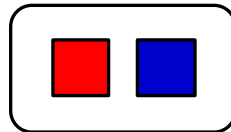
You Know that I Know That You Know...

We can define **beliefs** about other players' information sets!

Player 1's
information set:

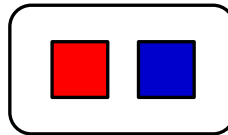


Player 2's
information set:

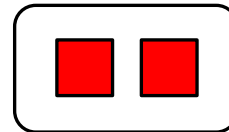
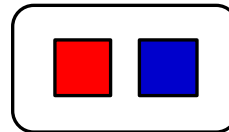


You Know that I Know That You Know...

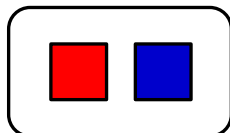
Player 1's
information set:



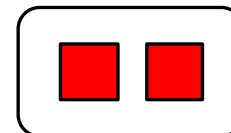
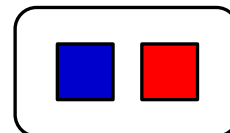
Player 2's
information set:



Player 2's belief about Player 1's knowledge

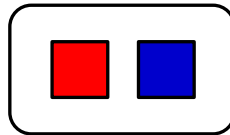


OR



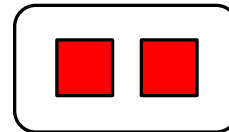
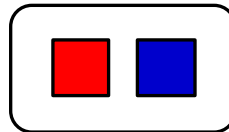
You Know that I Know That You Know...

Player 1's
information set:

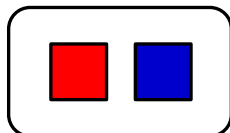


"Red"

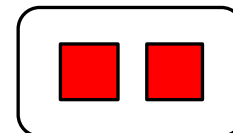
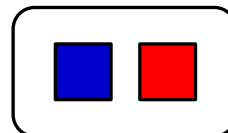
Player 2's
information set:



Player 2's belief about Player 1's knowledge

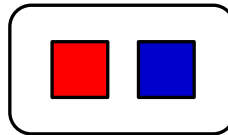


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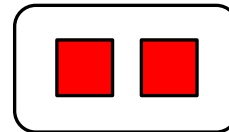
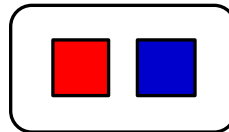
You Know that I Know That You Know...

Player 1's
information set:

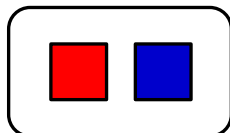


"Red"

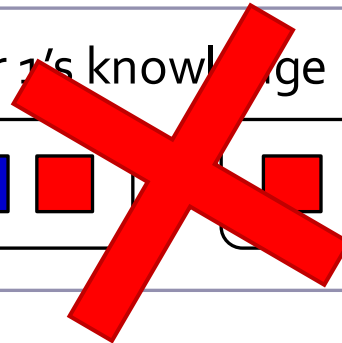
Player 2's
information set:



Player 2's belief about Player 1's knowledge

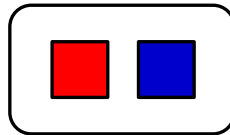


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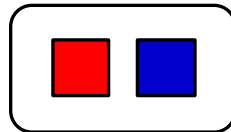
You Know that I Know That You Know...

Player 1's
information set:

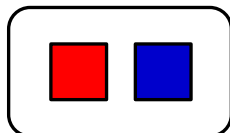


"Red"

Player 2's
information set:



Player 2's belief about Player 1's knowledge

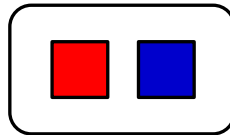


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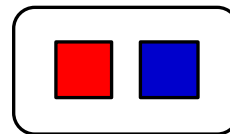


You Know that I Know That You Know...

Player 1's
information set:

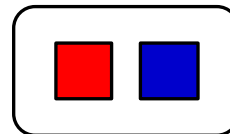


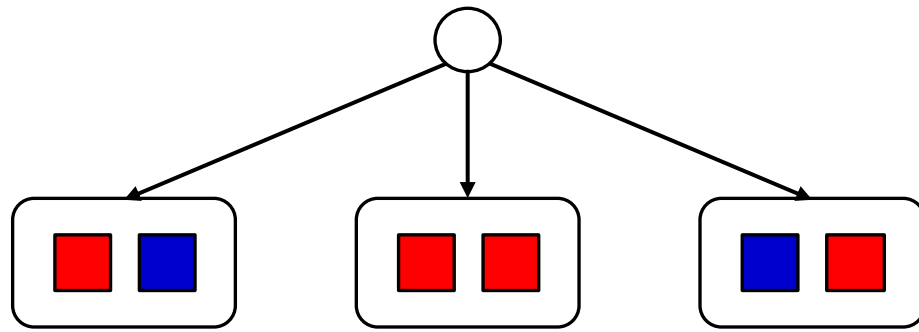
Player 2's
information set:

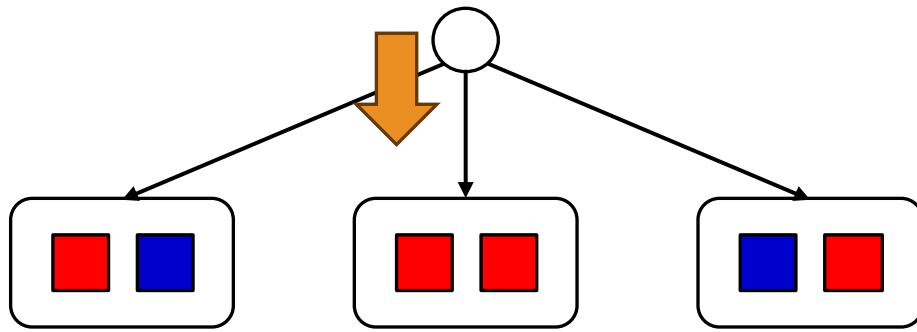


"Blue"

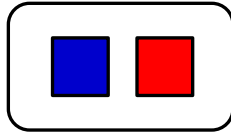
Player 2's belief about Player 1's knowledge



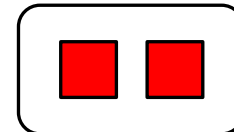
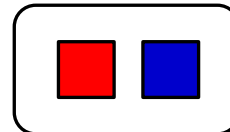




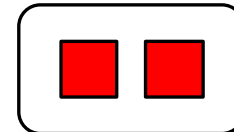
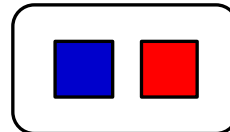
Player 1's knowledge about Player 2's knowledge



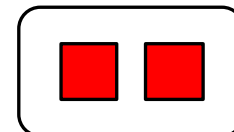
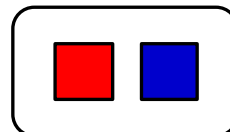
OR



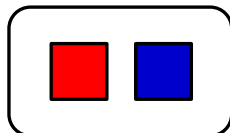
Player 1's
information set:



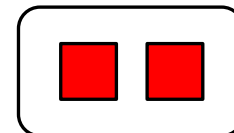
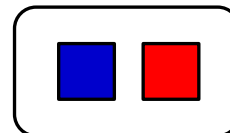
Player 2's
information set:



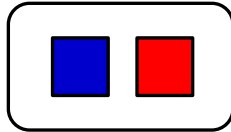
Player 2's knowledge about Player 1's knowledge



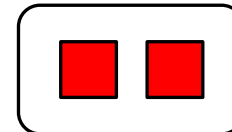
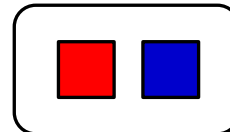
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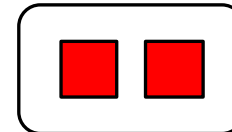
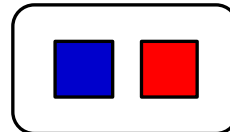
Player 1's knowledge about Player 2's knowledge



OR

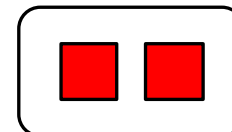
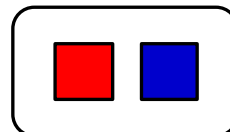


Player 1's
information set:

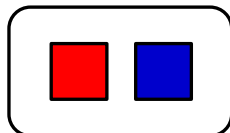


"Pass"

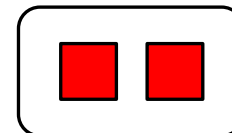
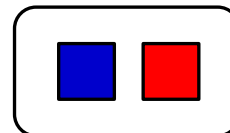
Player 2's
information set:



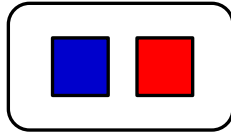
Player 2's knowledge about Player 1's knowledge



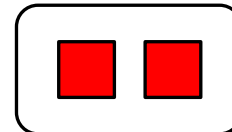
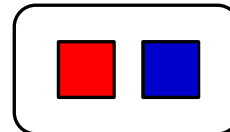
OR



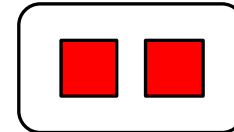
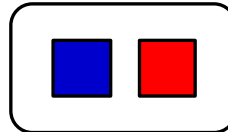
Player 1's knowledge about Player 2's knowledge



OR

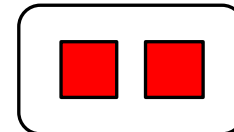
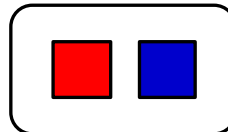


Player 1's
information set:

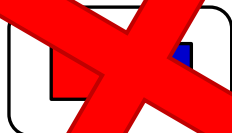


"Pass"

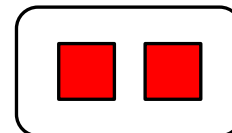
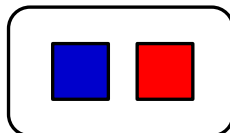
Player 2's
information set:



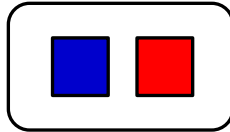
Player 2's knowledge about Player 1's knowledge



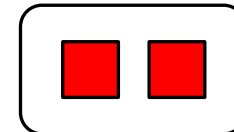
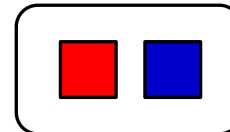
OR



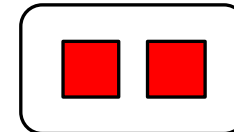
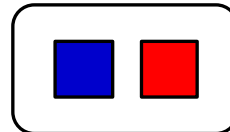
Player 1's knowledge about Player 2's knowledge



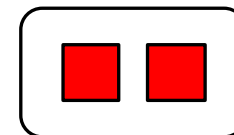
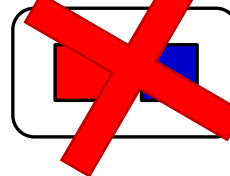
OR



Player 1's
information set:



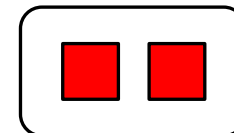
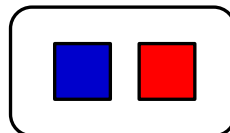
Player 2's
information set:



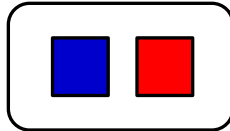
Player 2's knowledge about Player 1's knowledge



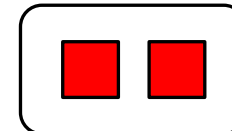
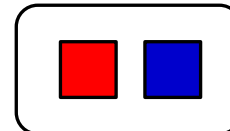
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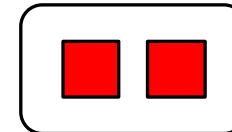
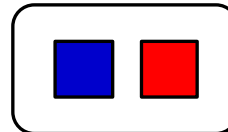
Player 1's knowledge about Player 2's knowledge



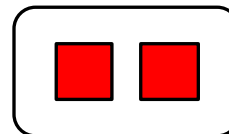
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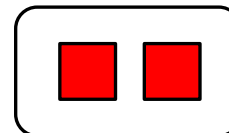
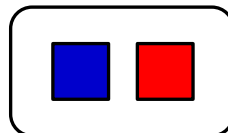
Player 1's
information set:



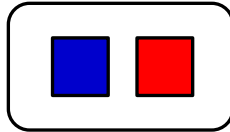
Player 2's
information set:



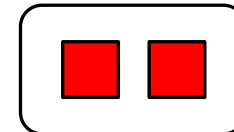
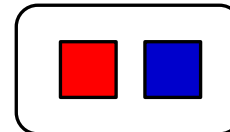
Player 2's knowledge about Player 1's knowledge



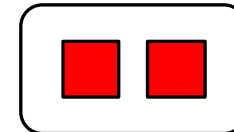
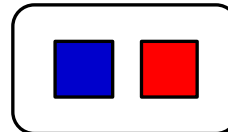
Player 1's knowledge about Player 2's knowledge



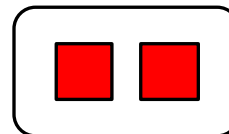
OR



Player 1's
information set:

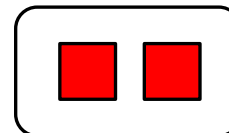
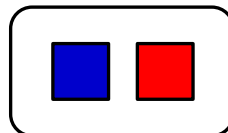


Player 2's
information set:



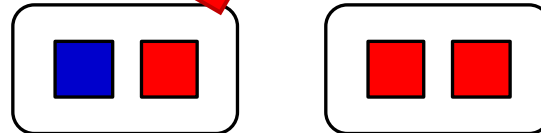
"Red"

Player 2's knowledge about Player 1's knowledge

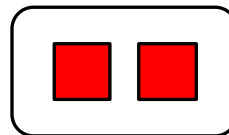




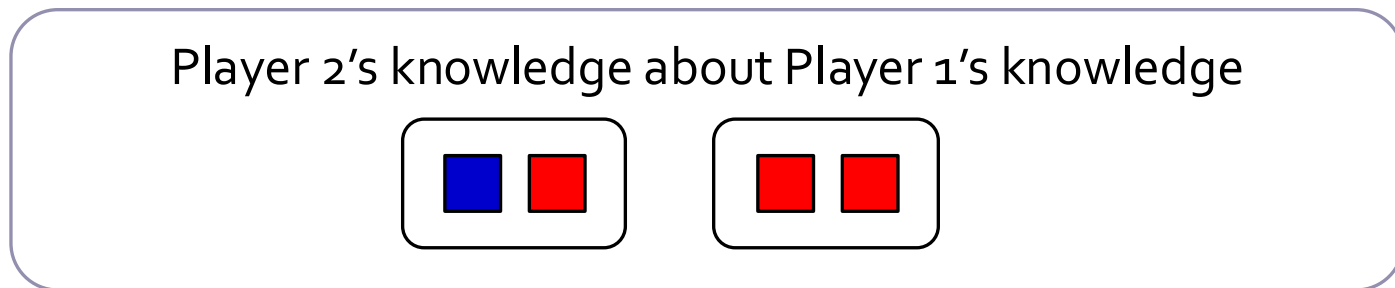
Player 1's
information set:



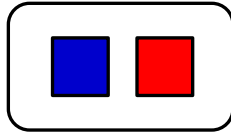
Player 2's
information set:



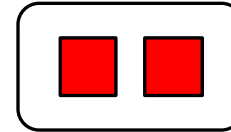
"Red"



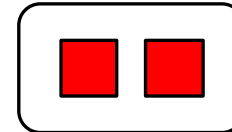
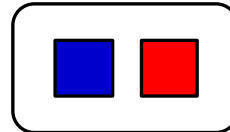
Player 1's knowledge about Player 2's knowledge



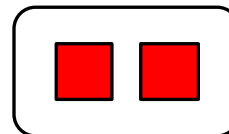
OR



Player 1's
information set:

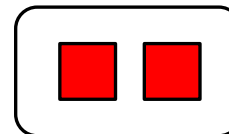
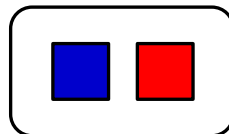


Player 2's
information set:

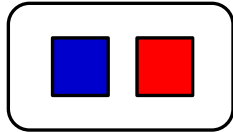


"Red"

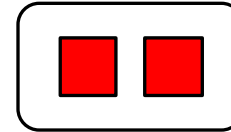
Player 2's knowledge about Player 1's knowledge



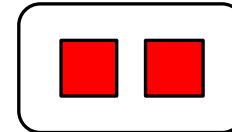
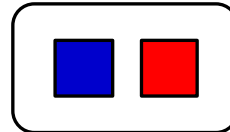
Player 1's knowledge about Player 2's knowledge



OR

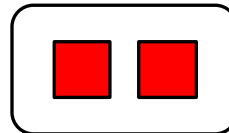


Player 1's
information set:

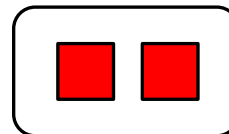
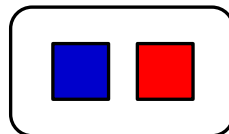


"Pass"

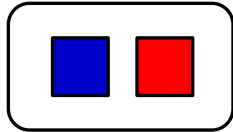
Player 2's
information set:



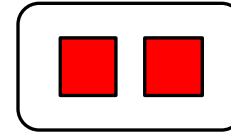
Player 2's knowledge about Player 1's knowledge



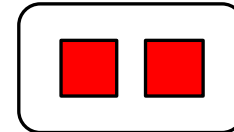
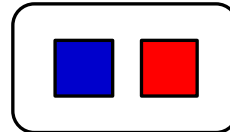
Player 1's knowledge about Player 2's knowledge



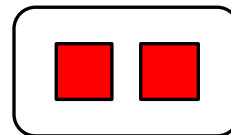
OR



Player 1's
information set:

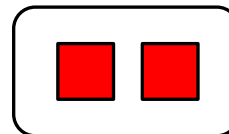
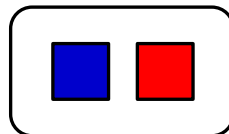


Player 2's
information set:

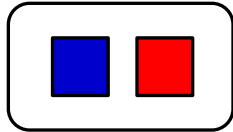


"Red"

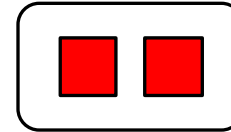
Player 2's knowledge about Player 1's knowledge



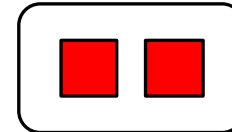
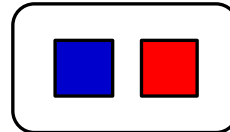
Player 1's knowledge about Player 2's knowledge



OR

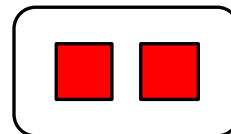


Player 1's
information set:



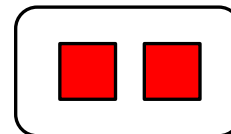
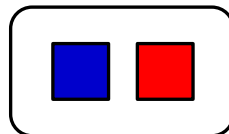
...

Player 2's
information set:



"Red"

Player 2's knowledge about Player 1's knowledge





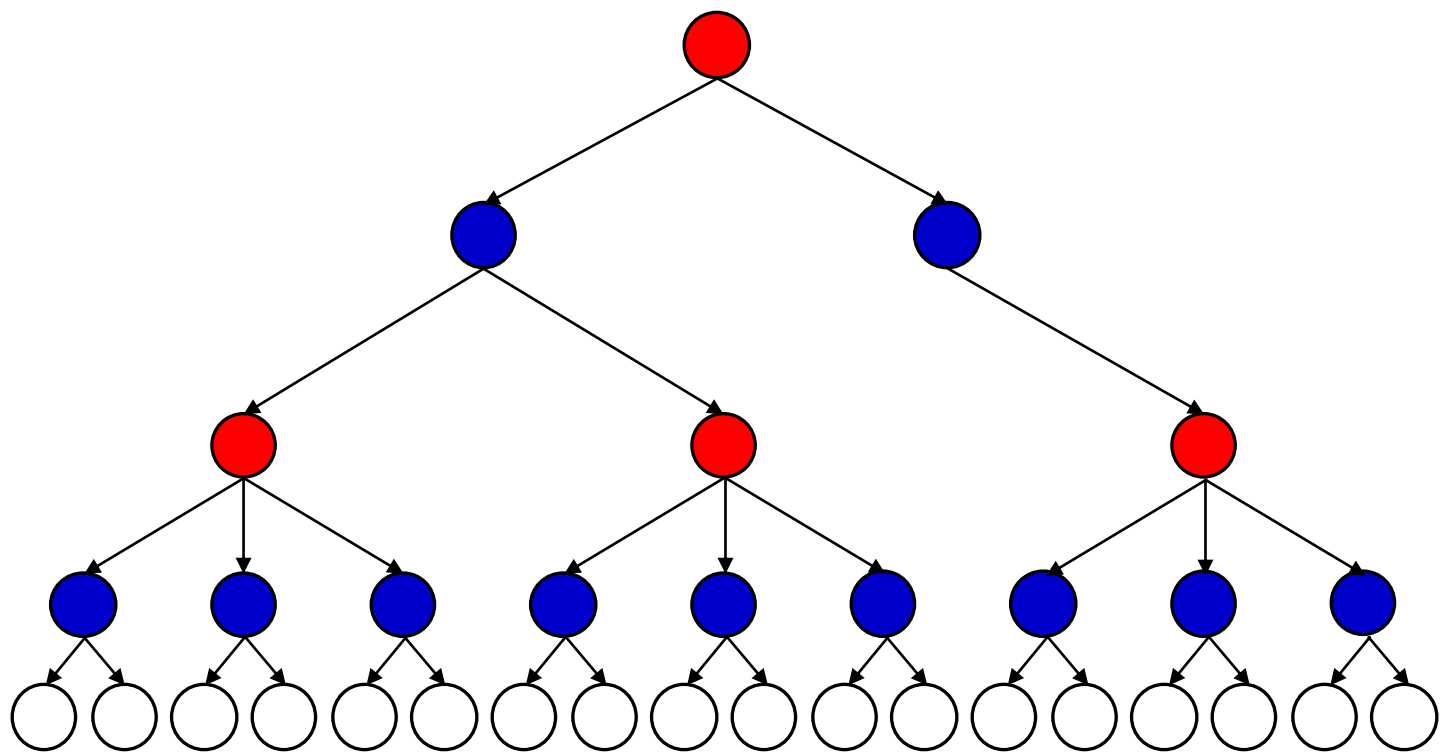
What happens if players announce their guesses simultaneously?

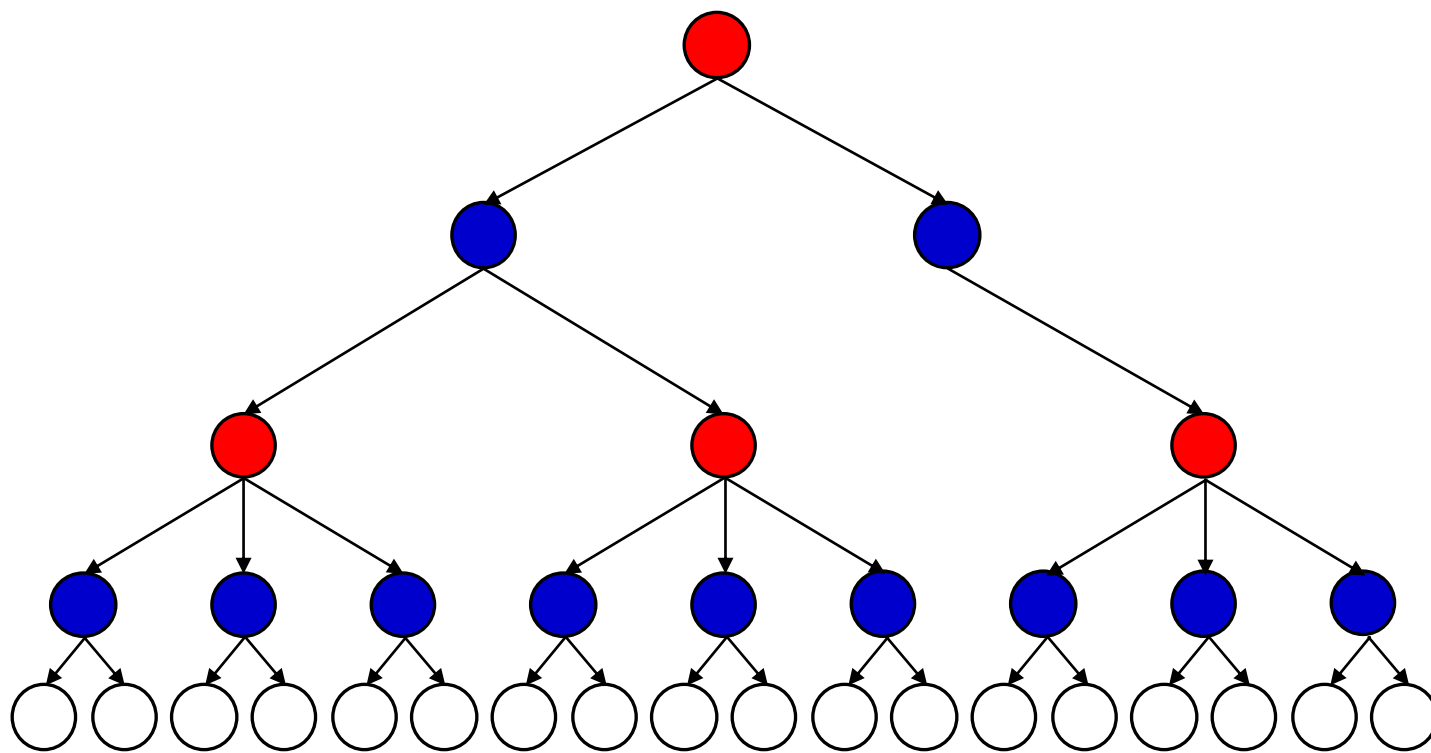
What happens if we have three players?



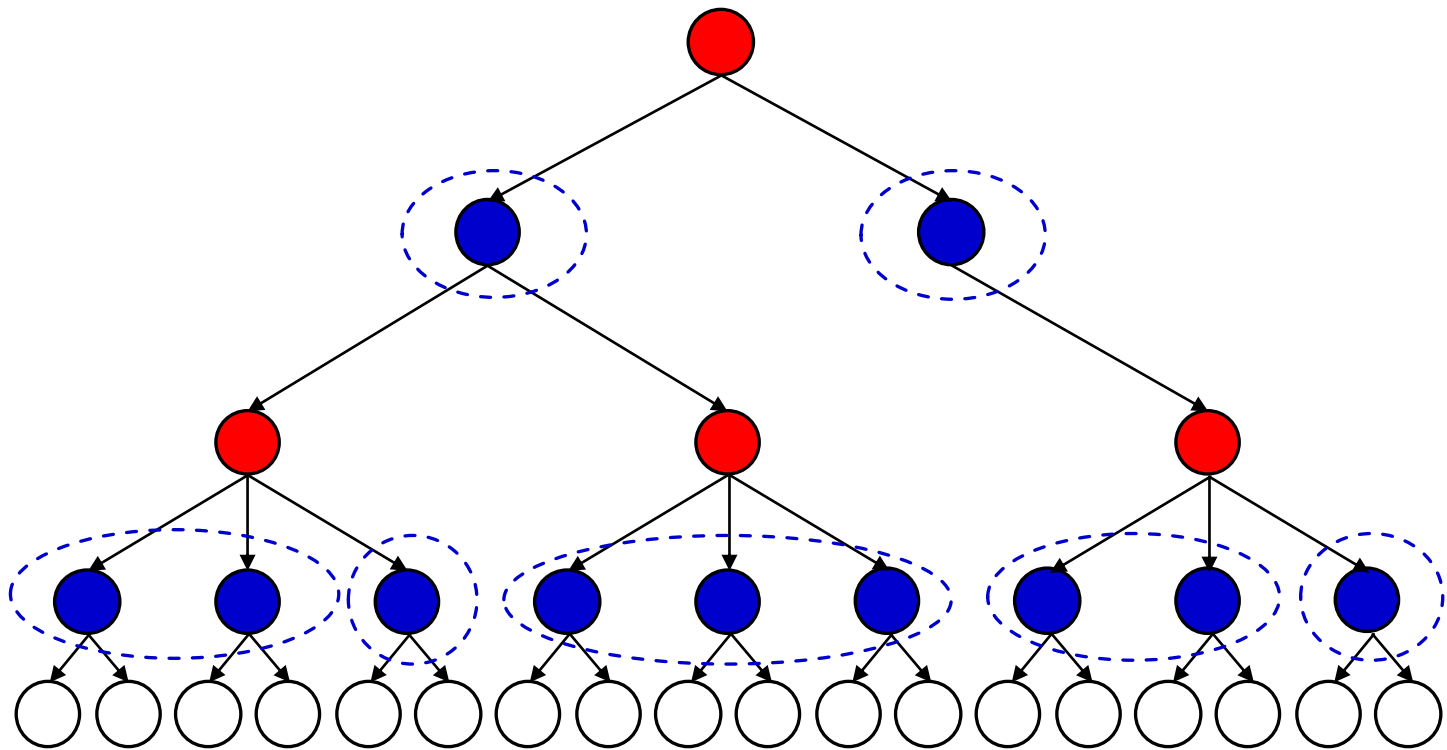
Beliefs and Information – a Formal Model

Zamir Maschler Solan's "Game Theory", Chapters 9 and 10



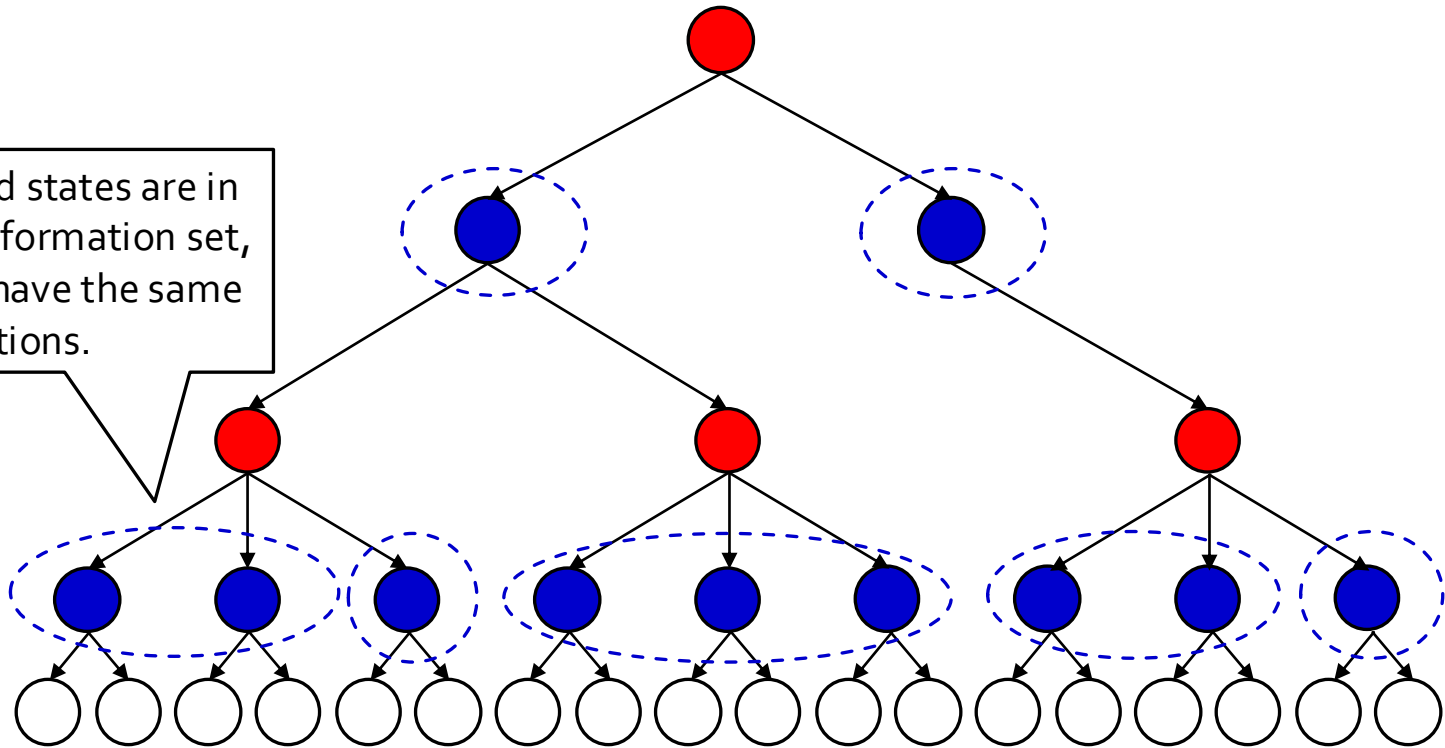


Player i 's belief space is some partition of the game states (more generally, a distribution over them).



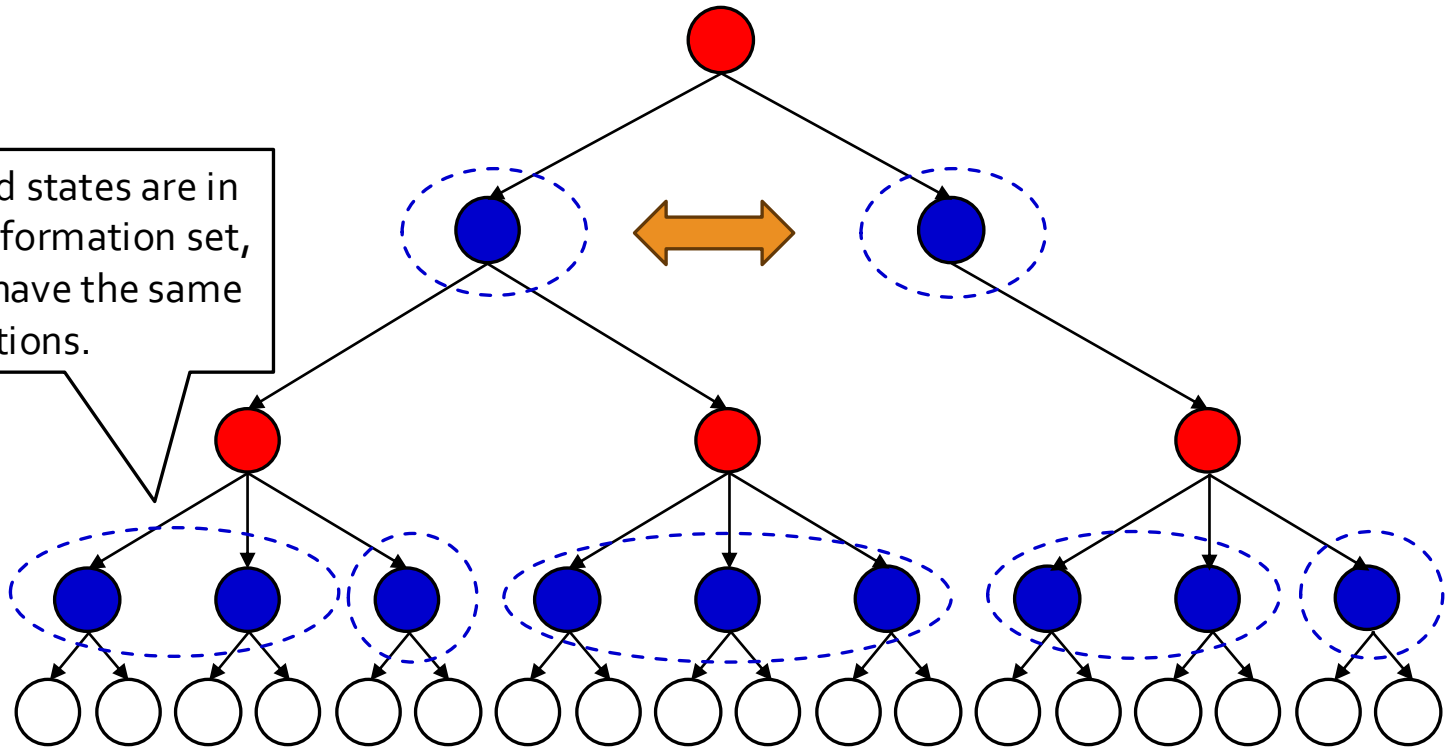
Player i 's belief space is some partition of the game states (more generally, a distribution over them).

If two world states are in the same information set, they must have the same actions.



Player i 's belief space is some partition of the game states (more generally, a distribution over them).

If two world states are in the same information set, they must have the same actions.



Player i 's knowledge space is some partition of the game states (more generally, a distribution over them).

Players operate in some *world*. The set of possible worlds is called Y .

Player i 's information $\mathcal{F}_i$ is a partition of Y .

$$\mathcal{F}_1 = \{\{\text{yellow globe}, \text{green globe}\}, \{\text{blue globe}, \text{purple globe}, \text{red globe}\}\}$$

$$\mathcal{F}_2 = \{\{\text{blue globe}, \text{purple globe}\}, \{\text{green globe}, \text{yellow globe}, \text{red globe}\}\}$$



Given a possible world $\omega \in Y$, we let $F_i(\omega)$ be the information set containing ω in player i 's information $\mathcal{F}_i$.



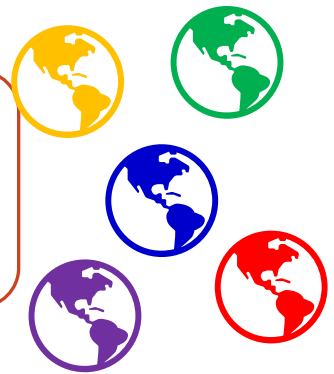
$$\omega = \text{yellow globe icon}$$

$$\mathcal{F}_1 = \{ \{ \text{yellow globe icon}, \text{green globe icon} \}, \{ \text{blue globe icon}, \text{purple globe icon}, \text{red globe icon} \} \}$$

$$\mathcal{F}_2 = \{ \{ \text{blue globe icon}, \text{purple globe icon} \}, \{ \text{green globe icon}, \text{yellow globe icon}, \text{red globe icon} \} \}$$

An *event* is some subset of worlds $A \subseteq Y$. We say that a player i *knows* A under the world state $\omega \in Y$ if

$$F_i(\omega) \subseteq A.$$



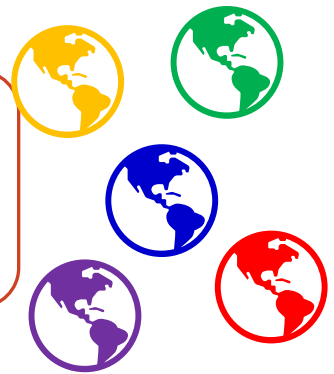
$$\omega = \text{yellow globe icon}$$

$$\mathcal{F}_1 = \{\{\text{yellow globe icon}, \text{green globe icon}\}, \{\text{blue globe icon}, \text{purple globe icon}, \text{red globe icon}\}\}$$

$$\mathcal{F}_2 = \{\{\text{blue globe icon}, \text{purple globe icon}\}, \{\text{green globe icon}, \text{yellow globe icon}, \text{red globe icon}\}\}$$

An *event* is some subset of worlds $A \subseteq Y$. We say that a player i *knows* A under the world state $\omega \in Y$ if

$$F_i(\omega) \subseteq A.$$



$$\omega = \text{purple globe} \quad A = \{\text{purple globe}, \text{blue globe}\}$$

$$\mathcal{F}_1 = \{\{\text{yellow globe}, \text{green globe}\}, \{\text{blue globe}, \text{purple globe}, \text{red globe}\}\}$$

$$\mathcal{F}_2 = \{\{\text{blue globe}, \text{purple globe}\}, \{\text{green globe}, \text{yellow globe}, \text{red globe}\}\}$$

Who knows A ?

Given an event $A \subseteq Y$, we let

$$K_i(A) = \{\omega \in Y : F_i(\omega) \subseteq A\}$$

$K_i(A) \subseteq Y$ is the set of all possible worlds where player i knows A .



$$A = \{\text{purple globe}, \text{blue globe}\}$$

$$\mathcal{F}_1 = \{\{\text{yellow globe}, \text{green globe}\}, \{\text{blue globe}, \text{purple globe}, \text{red globe}\}\}$$

$$\mathcal{F}_2 = \{\{\text{blue globe}, \text{purple globe}\}, \{\text{green globe}, \text{yellow globe}, \text{red globe}\}\}$$

When do players know A ?

We can define $K_j(K_i(A))$ as $\{\omega \in Y: F_j(\omega) \subseteq K_i(A)\}$.
This is the set of all worlds where player j knows that player i knows A .



$$A = \{\text{purple globe}, \text{blue globe}\}$$

$$\mathcal{F}_1 = \{\{\text{yellow globe}, \text{green globe}\}, \{\text{blue globe}, \text{purple globe}, \text{red globe}\}\}$$

$$\mathcal{F}_2 = \{\{\text{blue globe}, \text{purple globe}\}, \{\text{green globe}, \text{yellow globe}, \text{red globe}\}\}$$

When does Player 2 know
that Player 1 knows A ?

$K_1 K_3 K_2(A)$ = when does player 1 knows that player 3 knows that player 2 knows A ?

$K_2 K_5 K_1 K_2(A)$ = when does player 2 knows that player 5 knows that player 1 knows that player 2 knows A ?



An event $A \subseteq Y$ is *common knowledge* in the world $\omega \in Y$ if for any finite list of players $(i_1, \dots, i_\ell)$,

$$\omega \in K_{i_1} K_{i_2} \dots K_{i_\ell}(A)$$

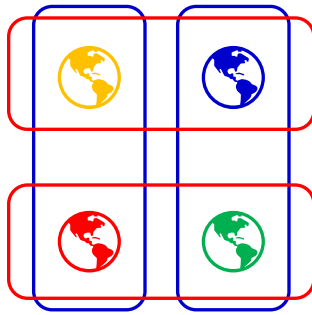
"Everyone knows A , everyone knows that everyone knows A , everyone knows that everyone knows that everyone knows A ..."

Adding priors: player i has a prior distribution over the possible worlds Y .

"given their behavior, it is unlikely that we are in a world where the opponent has a flush."

We say that player i knows the event $A \subseteq Y$ occurs in the world $\omega \in Y$ if they assign a probability of 1 to the event A given that we are in the world ω , i.e.

$$\Pr[A \mid F_i(\omega)] = 1 \Leftrightarrow F_i(\omega) \subseteq A$$



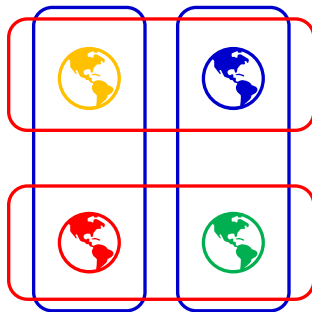
Common Prior:

$$\Pr[\text{Yellow Globe}] = \frac{1}{4}$$

$$\Pr[\text{Red Globe}] = \frac{1}{4}$$

$$\Pr[\text{Blue Globe}] = \frac{1}{3}$$

$$\Pr[\text{Green Globe}] = \frac{1}{6}$$



$$\omega = \text{yellow globe}$$

What are players' beliefs?

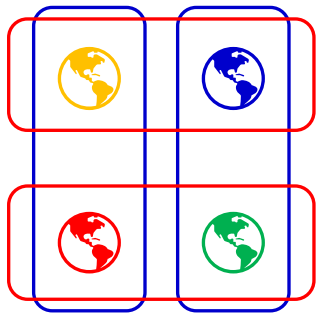
Common Prior:

$$\Pr[\text{yellow globe}] = \frac{1}{4}$$

$$\Pr[\text{red globe}] = \frac{1}{4}$$

$$\Pr[\text{blue globe}] = \frac{1}{3}$$

$$\Pr[\text{green globe}] = \frac{1}{6}$$



Common Prior:

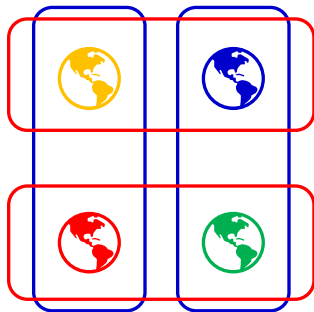
$$\begin{aligned}\Pr[\text{Yellow}] &= \frac{1}{4} \\ \Pr[\text{Red}] &= \frac{1}{4} \\ \Pr[\text{Blue}] &= \frac{1}{3} \\ \Pr[\text{Green}] &= \frac{1}{6}\end{aligned}$$

$$\omega = \text{Yellow}$$

What are players' beliefs?

Player 1:

$$\begin{aligned}\Pr[\text{Red} \mid \text{Yellow Red}] &= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = \frac{1}{2} \\ \Pr[\text{Yellow} \mid \text{Yellow Red}] &= \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{4}} = \frac{1}{2}\end{aligned}$$



Common Prior:

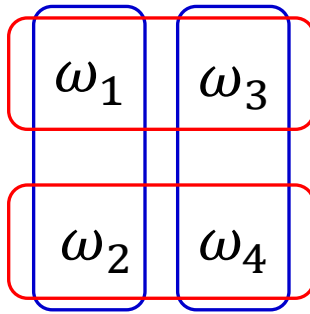
$$\begin{aligned}\Pr[\text{Yellow Globe}] &= \frac{1}{4} \\ \Pr[\text{Red Globe}] &= \frac{1}{4} \\ \Pr[\text{Blue Globe}] &= \frac{1}{3} \\ \Pr[\text{Green Globe}] &= \frac{1}{6}\end{aligned}$$

$$\omega = \text{Yellow Globe}$$

What are players' beliefs?

Player 2:

$$\begin{aligned}\Pr[\text{Blue Globe} \mid \text{Yellow Globe, Blue Globe}] &= \frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{4}} = \frac{4}{7} \\ \Pr[\text{Yellow Globe} \mid \text{Yellow Globe, Blue Globe}] &= \frac{\frac{1}{4}}{\frac{1}{3} + \frac{1}{4}} = \frac{3}{7}\end{aligned}$$

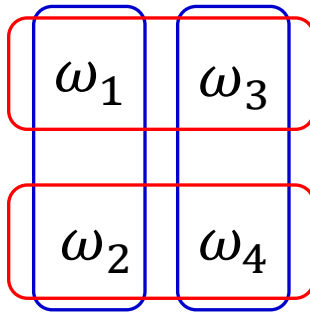


1st Order Beliefs

	ω_1	ω_2	ω_3	ω_4
Player 1	$\frac{1}{2}\omega_1 + \frac{1}{2}\omega_2$	$\frac{1}{2}\omega_1 + \frac{1}{2}\omega_2$	$\frac{2}{3}\omega_3 + \frac{1}{3}\omega_4$	$\frac{2}{3}\omega_3 + \frac{1}{3}\omega_4$
Player 2	$\frac{3}{7}\omega_1 + \frac{4}{7}\omega_3$	$\frac{3}{5}\omega_2 + \frac{2}{5}\omega_4$	$\frac{3}{7}\omega_1 + \frac{4}{7}\omega_3$	$\frac{3}{5}\omega_2 + \frac{2}{5}\omega_4$

Common Prior:

$$\begin{aligned}\Pr[\omega_1] &= \frac{1}{4} \\ \Pr[\omega_2] &= \frac{1}{4} \\ \Pr[\omega_3] &= \frac{1}{3} \\ \Pr[\omega_4] &= \frac{1}{6}\end{aligned}$$



Common Prior:

$$\begin{aligned}\Pr[\omega_1] &= \frac{1}{4} \\ \Pr[\omega_2] &= \frac{1}{4} \\ \Pr[\omega_3] &= \frac{1}{3} \\ \Pr[\omega_4] &= \frac{1}{6}\end{aligned}$$

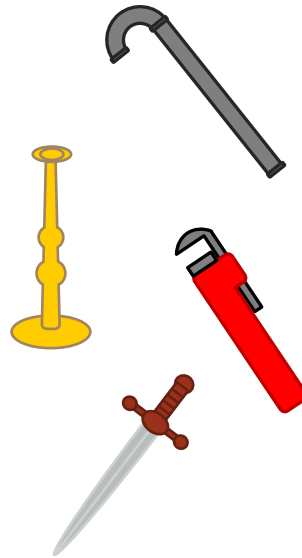
2nd Order Beliefs

If the world is ω_1 or ω_2 then player 1 believes that with prob. $\frac{1}{2}$, $\omega = \omega_1$ and Player 2's 1st order belief is $\frac{3}{7}\omega_1 + \frac{4}{7}\omega_3$.

With prob. $\frac{1}{2}$, $\omega = \omega_2$ and Player 2's 1st order belief is $\frac{3}{5}\omega_2 + \frac{2}{5}\omega_4$.

Cluedo:

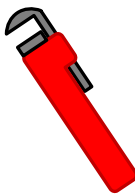
1. We place one person, weapon and room in an envelope.
2. We distribute the rest of the cards among the players.
3. On your turn, you may ask the players to reveal a person/weapon/room. If the player to your left has one of them, they must show them to you. If not, the next player does. **No one else sees the card shown.**
4. You may then opt to guess what's in the envelope.

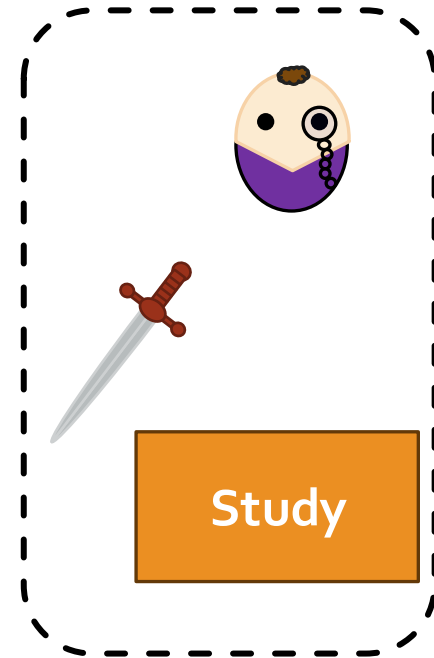


Kitchen

Dining
Room

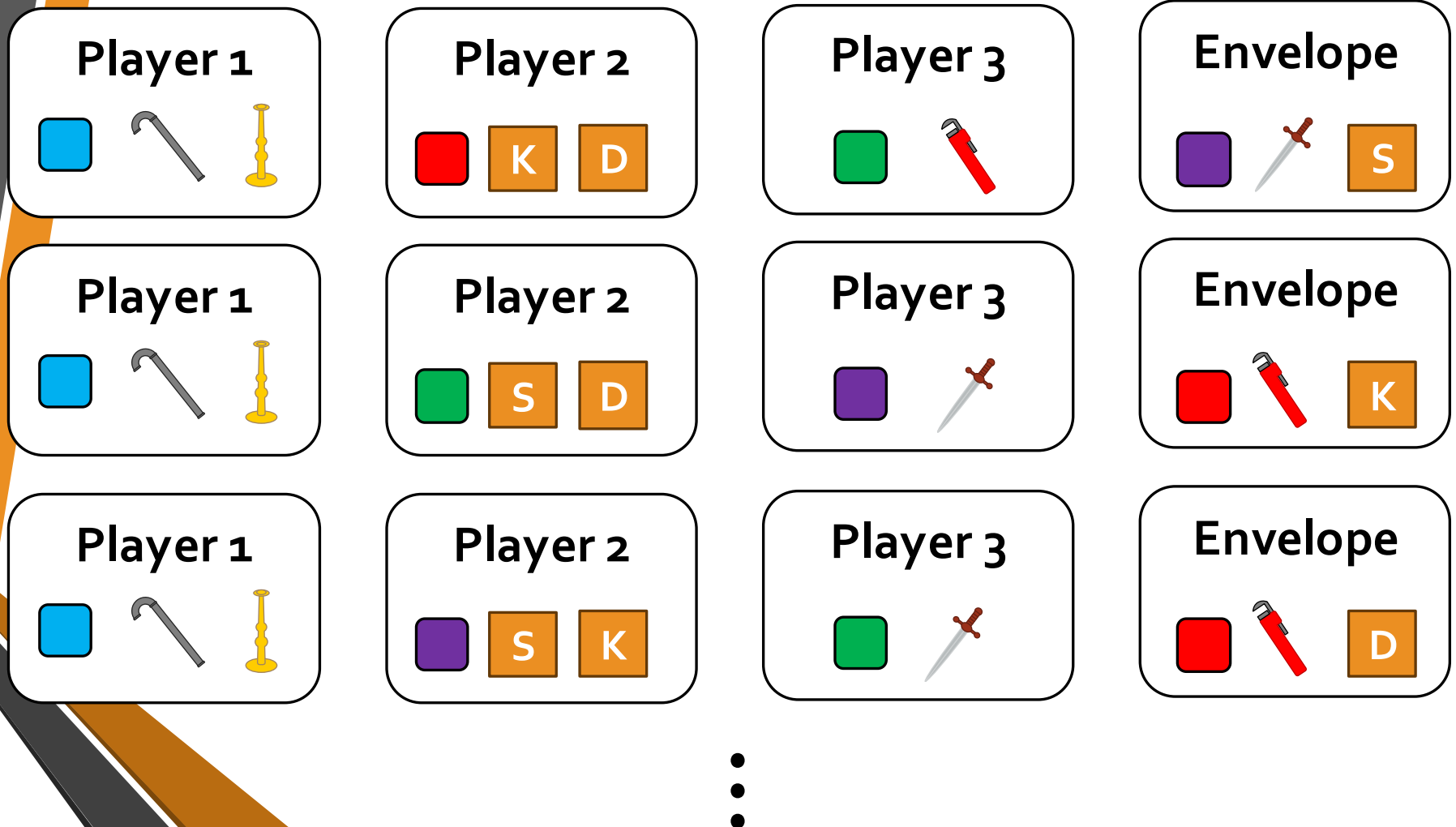
Study

Player 1	Player 2	Player 3
  	  	 



**What is the structure of players' information sets?
Can you list some possible worlds for player 1?**

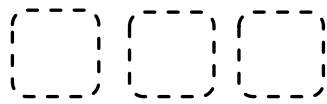
Information set of player 1: all possible legal partitions of the cards into cards in players' hands and the envelope.



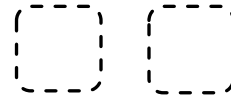
Player 1



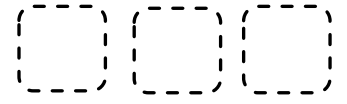
Player 2



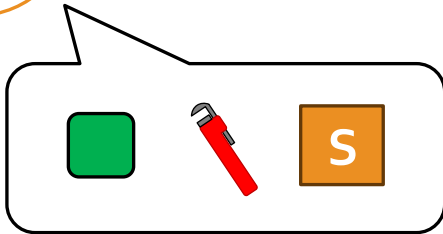
Player 3



Envelope



P₁



None

P₂

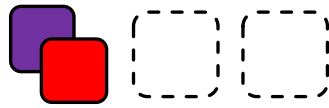


P₃

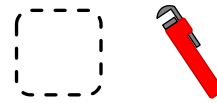
Player 1



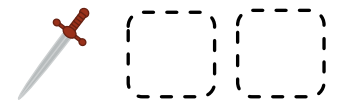
Player 2



Player 3



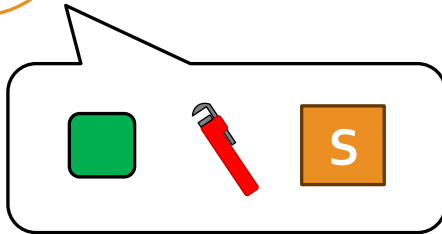
Envelope



P₁

<runs inference>

Player 3 has wrench $\Rightarrow$ Envelope has knife



None

P₂

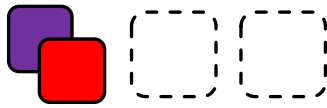


P₃

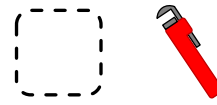
Player 1



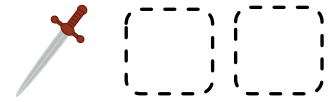
Player 2



Player 3



Envelope



P2



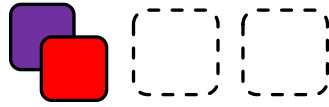
P3



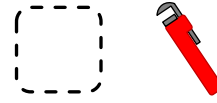
Player 1



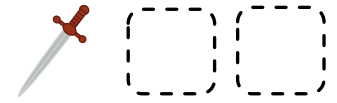
Player 2



Player 3



Envelope



P2



<runs inference>

P3 showed something $\Rightarrow$ P3 did
not show lead pipe $\Rightarrow$ P3 has
either kitchen or green

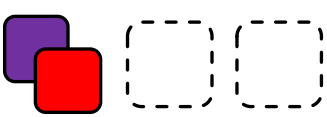


P3

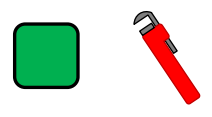
Player 1



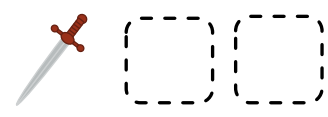
Player 2



Player 3



Envelope



Player 1



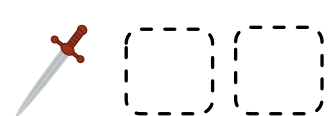
Player 2



Player 3



Envelope



P₁



P₂

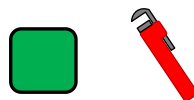
Player 1



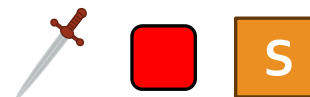
Player 2



Player 3



Envelope



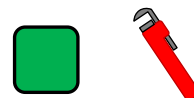
Player 1



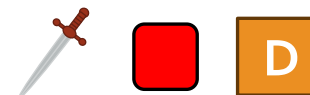
Player 2



Player 3



Envelope



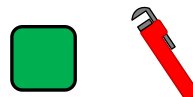
Player 1



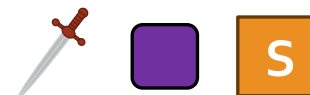
Player 2



Player 3



Envelope



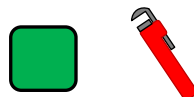
Player 1



Player 2

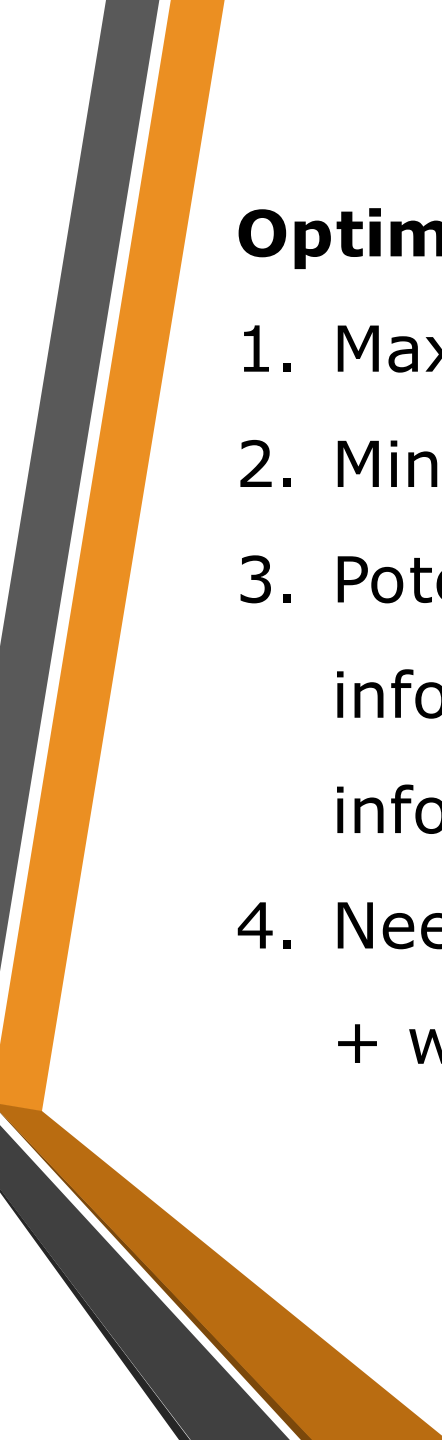


Player 3



Envelope





Optimal clue selection:

1. Maximizing information I gain.
2. Minimizing information gained by others.
3. Potentially select clue that offers less information if it prevents others from obtaining information.
4. Need to model what I know + what others know + what others know I know ...